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On families of sets without k pairwise disjoint members

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ABSTRACT

For integers $n \geq k \geq 2$ let $m(n,k)$ denote the maximum size of a family of subsets of an n-set without k pairwise disjoint members. In an important classical paper Kleitman determined $m(n,k)$ for $n \equiv 0$ or $-1 \pmod{k}$. We present a simple short proof of this result and prove the uniqueness of the optimal families as well.

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1. Introduction

Let $X = \{1, 2, \ldots, n\}$ be the standard n-element set and 2^X its power set. For $0 \leq i \leq n$ let $\binom{X}{i} \subset 2^X$ denote the collection of all i-subsets of X. Subsets of 2^X are called families. For an integer $k \geq 2$ a family $\mathcal{F} \subset 2^X$ is called *k-dependent* if it contains no k pairwise disjoint members.

Let us define $m(n,k) = \max\{|\mathcal{F}| : \mathcal{F} \subset 2^X, \mathcal{F} \text{ is } k\text{-dependent}\}$. If $\mathcal{F}$ is 2-dependent and $F \in \mathcal{F}$ then the complementary set $X \setminus F$ cannot be in $\mathcal{F}$. Noting this fact Erdős–Ko–Rado [1] proved $m(n,2) = 2^{n-1}$.

Let us recall the following classical result of Kleitman [6].

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Kleitman Theorem *([6]). For all $k \geq 2$, $s \geq 1$ one has:*

$$m(ks-1,k) = \sum_{i \geq s} \binom{ks-1}{i}, \tag{1.1}$$

$$m(ks,k) = \binom{ks-1}{s} + \sum_{i > s} \binom{ks}{i}. \tag{1.2}$$

Providing families attaining equality is easy. For $n = ks - 1$ define $\mathcal{K}(n,s) = \{K \subset X : |K| \geq s\}$. For $n = ks$ fix a subset $Y \in \binom{X}{n-1}$ and define

$$\mathcal{K}(n,s) = \binom{Y}{s} \cup \{K : K \subset X, |K| > s\}.$$

Noting that $\mathcal{K}(ks,k) = \{K \subset X : |K \cap Y| \geq s\}$ one deduces $|\mathcal{K}(ks,k)| = 2|\mathcal{K}(ks-1,k)|$.

Kleitman notes that whenever $\mathcal{G} \subset 2^Y$ is k-dependent, its "double" $\mathcal{D} = \{D \subset X : D \cap Y \in \mathcal{G}\}$ is k-dependent as well. This proves

$$m(n+1,k) \geq 2m(n,k). \tag{1.3}$$

Thereby (1.1) is a consequence of (1.2).

The proof of Kleitman is very clever but not simple. His paper is short because most computations are only sketched. The purpose of the present note is to provide a very short and hopefully illuminating proof of (1.2).

Throughout the paper we let $k \geq 3$ be fixed. Note that if $\mathcal{F} \subset 2^X$ is k-dependent then its closure, $\mathcal{F}^* = \{G \subset X : \exists F \in \mathcal{F}, F \subset G\}$ is k-dependent as well. Therefore we are going to assume always that $\mathcal{F}$ is an *up-set* (i.e., $F \in \mathcal{F}$ and $F \subset G \subset X$ imply $G \in \mathcal{F}$). Equivalently, if $H \notin \mathcal{F}$ and $G \subset H$ then $G \notin \mathcal{F}$ as well. In particular $\emptyset \notin \mathcal{F}$.

Following Kleitman let $y_j = y_j(\mathcal{F})$ denote the number j-subsets of X that are *not* members of $\mathcal{F}$.

Thus (1.2) is equivalent to

$$\sum_{0 \leq j \leq n} y_j(\mathcal{F}) \geq \binom{ks-1}{s-1} + \sum_{0 \leq i < s} \binom{ks}{i}. \tag{1.4}$$

Our proof of (1.4) is based on simple averaging and Lemma 1 below. To state the lemma we need to define an auxiliary family $\mathcal{A}$.

Let $n = ks$ and fix a partition $X = B_1 \cup \ldots \cup B_k$, where $|B_1| = \ldots = |B_k| = s$.

For $1 \leq i \leq k$ and $0 < j < s$ let $A_j(i)$ be an arbitrary j-subset of B_i.

For every pair of *distinct* numbers $1 \leq i \neq i' \leq k$ and $1 \leq j < k$ define $C_{2s-j}(i,i') = (B_i \cup B_{i'}) \setminus A_j(i)$ and note that $B_i \cup B_{i'}$ is the disjoint union of $A_j(i)$ and $C_{2s-j}(i,i')$.

Now let $\mathcal{A}$ be the collection of all these sets. Note that

$$\left|\mathcal{A} \cap \binom{X}{j}\right| = \begin{cases} 1 & \text{for } j = 0, \\ k & \text{for } 1 \le j \le s, \\ k(k-1) & \text{for } s < j < 2s, \\ 0 & \text{otherwise.} \end{cases} \tag{1.5}$$

Lemma 1. *Suppose that $\mathcal{F} \subset 2^X$ is a k-dependent up-set. Let z_j denote the number of j-sets in $\mathcal{A}$ that are* not *in $\mathcal{F}$. Then* (i) *or* (ii) *holds.*

(i) $z_0 = 1$ *and* $z_j \ge 2$ *for all* $1 \le j \le s$*;*
(ii) $z_0 = z_s = 1$ *and* $z_j + z_{2s-j} \ge k$ *for all* $1 \le j < s$*.*

Proof. Since $\mathcal{F}$ is k-dependent, at least one of the sets $B_1, \dots, B_k \in \mathcal{A}$ is missing from $\mathcal{F}$. If for $1 \le i < i' \le k$ we have $B_i \notin \mathcal{F}$, $B_{i'} \notin \mathcal{F}$. Then $A_j(i) \notin \mathcal{F}$ and $A_j(i') \notin \mathcal{F}$ hold as well and the validity of (i) follows.

Thus we may assume by symmetry that $B_1, \dots, B_{k-1} \in \mathcal{F}$ but $B_k \notin \mathcal{F}$. Again, $A_j(k) \notin \mathcal{F}$ for $0 \le j < s$ because $\mathcal{F}$ is an up-set. To conclude the proof we show $z_j + z_{2s-j} \ge k$ $(1 \le j < s)$. It is very simple. Just note that for every $i < k$ the k sets $A_j(i)$, $C_{2s-j}(i,k)$ and B_t, $1 \le t < k$, $t \ne i$ form a partition of X. Consequently, either $A_j(i)$ or $C_{2s-j}(i,k)$ is missing from $\mathcal{F}$. Since we have $k-1$ choices for i and we know that $A_j(k) \notin \mathcal{F}$,

$$z_j + z_{2s-j} \ge k \quad \text{follows.} \quad \square$$

The above lemma will be the only combinatorial statement used in the proof of (1.4) which we establish in the following, slightly stronger form

$$\sum_{0 \le j < 2s} y_j(\mathcal{F}) \ge \binom{ks-1}{s-1} + \sum_{0 \le j < s} \binom{ks}{j}. \tag{1.6}$$

We should mention that Kleitman proved (1.6) in a stronger form. Namely, summing only for $0 \le j \le s+1$.

2. Some simple inequalities

The aim of this section is to prove inequalities (2.2) and (2.4) that will be used in the sequel.

First note the following:

$$\binom{ks}{a} > (k-1)\binom{ks}{a-1} \quad \text{is equivalent to} \quad a \le s. \tag{2.1}$$

Indeed, $\binom{ks}{a} / \binom{ks}{a-1} = (ks - a + 1)/a$ implying (2.1). Using (2.1) we deduce

$$2 \sum_{0 \le j < s} \binom{ks}{s-j} > \binom{ks}{s} + k \sum_{1 \le j < s} \binom{ks}{s-j}. \tag{2.2}$$

Rearranging leads to

$$\binom{ks}{s} > (k-2) \sum_{1 \le j < s} \binom{ks}{s-j}. \tag{2.3}$$

However, by (2.1), $\binom{ks}{s-j} < (k-1)^{-j}\binom{ks}{s}$ and (2.3) follows from

$$(k-2) \sum_{j \ge 1} (k-1)^{-j} = 1.$$

From (2.1) the following inequality is immediate:

$$\binom{ks}{2s-j} > (k-1)\binom{ks}{j} \quad \text{for} \quad 1 \le j < s. \tag{2.4}$$

3. The proof of (1.6)

Let π be an arbitrary permutation of X. Let $z_j(\pi)$ denote the number of j-sets $0 \le j < 2s$ in $\pi(\mathcal{A}) = \{\pi(A) : A \in \mathcal{A}\}$ that are *not* in $\mathcal{F}$.

Proposition 1. *For all $n!$ permutations π one has*

$$z_s(\pi)\binom{n}{s} + \sum_{0<j<s} \left(z_j(\pi) + z_{2s-j}(\pi)\right)\binom{n}{j} \ge \binom{n}{s} + k \sum_{0<j<s} \binom{n}{j}. \tag{3.1}$$

Proof. Applying Lemma 1 to the family $\pi(\mathcal{A})$ we distinguish two cases. If (i) holds then $z_j(\pi) \ge 2$ for all $1 \le j \le s$ and (3.1) follows from (2.2).

In case (ii) it is immediate directly. □

Now note that for a fixed j-set $A \in \mathcal{A}$ and a fixed j-element non-member F of $\mathcal{F}$ there are $j!(n-j)!$ permutations π with $\pi(A) = F$. Also, $j!(n-j)!\binom{n}{j} = n!$. Consequently, averaging (3.1) over all $n!$ permutations yields

$$ky_s + \sum_{0<j<s} \left(ky_j + k(k-1)\frac{\binom{n}{j}}{\binom{n}{2s-j}} y_{2s-j} \right) \ge \binom{n}{s} + k \sum_{0<j\le s} \binom{n}{j}. \tag{3.2}$$

Since by (2.4) the coefficient of y_{2s-j} is less than k, dividing (3.2) by k and adding $y_0 = 1$ gives

$$\sum_{0\le j<2s} y_j \ge \frac{1}{k}\binom{n}{s} + \sum_{0\le j<s}\binom{n}{j} \quad \text{as desired.} \quad \square$$

Recall that $k \ge 3$ and let us analyze the case of equality. In view of (2.2), (ii) must hold in Lemma 1 for every permutation π. That is, for every equipartition $X = D_1 \cup \ldots \cup D_k$, $|D_i| = s$ *exactly* one of the sets D_i is a non-member of $\mathcal{F}$. In [5] we proved in a more general setting that this happens if and only if $\mathcal{F} \cap \binom{X}{s} = \binom{Y}{s}$ for some $Y \in \binom{X}{n-1}$.

Also, (2.4) implies that $z_{2s-j}(\pi) = 0$ for all $1 \le j < s$. Thus $\binom{X}{2s-j} \subset \mathcal{F}$ for all $1 \le j < s$. Using that $\mathcal{F}$ is an up-set, $\mathcal{F} = \mathcal{K}(n,k)$ follows. Let us mention that this characterization of the families meeting the bound (1.2) is new.

The uniqueness for the case $n = ks - 1$ follows also. Namely, if $\mathcal{G} \subset 2^Y$ is meeting the bound (1.1) on $Y \in \binom{X}{ks-1}$ then its double $\mathcal{D}(\mathcal{G})$ is meeting the bound (1.2) on X (cf. the argument leading to (1.3)).

4. Concluding remarks

To determine $m(n,k)$ for $n \not\equiv 0$ or $-1 \pmod k$ is a notoriously difficult problem. In his Ph. D. dissertation Quinn [7] gave a long proof for

$$m(3s+1,3) = \binom{3s}{s-1} + \sum_{j>s}\binom{3s+1}{j}.$$

However, his proof was never published.

Recently for $n = ks + k - 2$ Kupavskii and the author [2], [4] proved

$$m(ks+k-2,k) = \binom{n-1}{s-1} + \sum_{j>s}\binom{n}{j}$$

for arbitrary $k \ge 3$. In [3] they established a best possible multi-family version of Kleitman's result for all n and k that implies both (1.1) and (1.2). The proofs in both papers are rather complicated.

One can modify the proof that we presented to provide a proof of the multi-family version, that is, to determine the maximum of $|\mathcal{F}_1| + \ldots + |\mathcal{F}_k|$ over all families $\mathcal{F}_1, \ldots, \mathcal{F}_k \subset 2^{[n]}$ containing no members $F_1 \in \mathcal{F}_1, \ldots, F_k \in \mathcal{F}_k$ that are pairwise disjoint, $n \equiv 0$ or $-1 \pmod k$. However, the case of the remaining congruence classes appears to be more difficult.

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References

[1] P. Erdős, C. Ko, R. Rado, Intersection theorems for systems of finite sets, Quart. J. Math. Oxf. Second Ser. 12 (1961) 313–320.
[2] P. Frankl, A. Kupavskii, Families with no s pairwise disjoint sets, J. Lond. Math. Soc. 95 (2017) 875–894.
[3] P. Frankl, A. Kupavskii, Two problems of P. Erdős on matchings in set families, arXiv:1607.06126, submitted for publication.
[4] P. Frankl, A. Kupavskii, Families of sets with no matchings of sizes 3 and 4, submitted for publication.
[5] P. Frankl, The shifting technique in extremal set theory, in: Surveys in Combinatorics 1987, in: London Math. Soc. Lecture Note Ser., vol. 123, Cambridge Univ. Press, Cambridge, 1987, pp. 81–110.
[6] D.J. Kleitman, Maximal number of subsets of a finite set no k of which are pairwise disjoint, J. Combin. Theory 5 (1968) 157–163.
[7] F. Quinn, PhD Thesis, Massachusetts Institute of Technology, 1986.